Bharat Goel

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2025	9.00
Shaheed Rajpal DAV Public School Dayanand Vihar	Class XII (CBSE)	2020	87.00%
Shaheed Rajpal DAV Public School Dayanand Vihar	Class X (CBSE)	2018	85.00%

WORK EXPERIENCE

POLESTAR LLP, ANALYST(JUNE 2024 - MAY 2025)

- Developed APIs to connect backend for the P.AI portal, an internal project for Polestar to the database.
- Contributed to the creation of the GenAI-powered P.AI chatbot, enhancing user interaction.
- Led POC development for multiple clients, showcasing innovative solutions. Performed data modeling and prompt engineering to optimize chatbot performance.

SKILLS

Languages: Python, C++, SQL

Core CS: Data Structures & Algorithms, Object-Oriented Programming, Complexity Analysis, DBMS

Mathematics: Probability, Statistics, Linear Algebra, Calculus

Machine Learning: Linear Regression, Logistic Regression, Naive Bayes, Support Vector Machines, Model Evaluation

Deep Learning: Neural Networks, MLPs, Backpropagation, Gradient Descent, Activation Functions, Loss Functions, Regularization, PyTorch

Libraries: NumPy, Pandas, Matplotlib, Scikit-learn

Tools: Git, GitHub, Docker, Jupyter Notebook, VS Code

RELEVANT COURSES

Machine Learning, Probability and Stochastic Processes-Theory and Applications, Linear Algebra, Statistical Inference, Deep Learning, Convex Optimization

PROJECTS

Neural Movie Recommendation System | Python, PyTorch, FastAPI, Streamlit, Docker | GitHub

- Built a two-stage neural recommendation system on the **MovieLens 1M** dataset, using a **Two-Tower architecture** for candidate retrieval followed by a **neural ranking model** for reranking.
- Implemented user/movie embeddings, **negative sampling**, chronological per-user train/validation/test splits, and dot-product retrieval; evaluated using **Recall@10/50/100, NDCG@10, and MRR**.
- Compared against **Popularity, Matrix Factorization, and BPR** baselines; Two-Tower + Ranking achieved **0.0435 Recall@10, 0.1603 Recall@50, 0.2561 Recall@100, 0.0646 NDCG@10, and 0.1544 MRR**.
- Developed an end-to-end inference pipeline and deployed the recommender through a **FastAPI REST API** and interactive **Streamlit application**, with **Docker** containerization for reproducible deployment.

Production RAG & Evaluation System | *Python, LangChain, Chroma, RAGAS, LangSmith, FastAPI, Docker*

- Built an end-to-end **RAG pipeline** over **20 AI/ML research papers** using **dense retrieval, BM25, Reciprocal Rank Fusion (RRF), BGE cross-encoder reranking**, and **Claude-based grounded generation**.

- Designed a **50-question evaluation framework** with **RAGAS and LangSmith**; improved **context precision from 0.275 to 0.687** and **context recall from 0.400 to 0.900** through systematic retrieval and reranking experiments.
- Deployed a production-style **FastAPI + Streamlit application** with **source/page attribution, LangSmith observability, Pytest testing, Ruff linting, Docker containerization**, and **GitHub Actions CI/CD**.

Real-Time Fraud Detection System | Python, XGBoost, Kafka, FastAPI, Streamlit, Docker

- **Built** an end-to-end fraud detection pipeline using **Optuna-tuned XGBoost**, temporal data splitting, feature engineering, and validation-based decision threshold optimization.
- **Deployed** real-time inference using **Apache Kafka and FastAPI**, processing streaming transactions and generating automated **ALLOW/BLOCK decisions and fraud alerts**.
- **Implemented PSI-based drift monitoring** and a **Streamlit dashboard**, then **Dockerized** the Kafka, API, producer, and consumer services using Docker Compose.

CURRENT RESEARCH

Regret Minimization in Markov Chain Multi-Arm Bandits

Currently implementing and analyzing Multi-Armed Bandit algorithms (UCB, Thompson Sampling, and EXP3) for stochastic and adversarial setting with a focus on regret analysis and sequential decision-making under uncertainty. Future aim is to research and analyze new algorithms for Markov setting.

POSITION OF RESPONSIBILITIES

Teaching Assistant (TA) for Concentration Inequalities (AI2200), IIT Hyderabad | Aug 2026 - Sep 2026

Teaching Assistant for the graduate-level course on concentration inequalities and their applications in probability and machine learning.